

OceanSaksham: An Open-Source Architecture for Credibility-Aware Triage and Audit Logging in Crowdsourced Coastal Hazard Management

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Abstract—Crowdsourced citizen reporting is an important mechanism for gathering hyper-local situational awareness during coastal emergencies such as storm surges, high wave breaches, tidal inundation, and shoreline erosion. However, emergency operation centres and disaster management agencies frequently encounter operational bottlenecks caused by unverified, noisy, duplicate, or uncalibrated submissions. In this paper, we present the system design, multi-signal feature architecture, and reproducible synthetic benchmark evaluation of OceanSaksham, an open-source platform for coastal hazard reporting featuring an explainable, human-in-the-loop heuristic triage layer and cryptographically hash-chained audit logging, alongside an offline machine learning benchmark harness for future data-driven scoring.

The proposed triage engine formulates 19 features across seven modalities: geospatial GPS precision, camera EXIF metadata cross-validation, temporal submission latency, multilingual Indic keyword alignment (across nine coastal Indian languages), exact-match cryptographic media hashing (SHA-256), spatial-temporal neighborhood corroboration, and model-derived sea-state cross-validation via Open-Meteo marine data. To evaluate triage mechanics under controlled conditions, we formulate a 5-seed parameterized synthetic benchmark ($N = 1,000$ per seed) evaluated under a region-held-out spatial split (training on West Coast regions, evaluating on East Coast regions). Across five independent dataset generations, regularized logistic regression achieves $F_1 = 0.946 \pm 0.004$ (AUC-ROC = 0.988 ± 0.001), random forest achieves $F_1 = 0.938 \pm 0.009$ (AUC-ROC = 0.987 ± 0.002), and gradient-boosted decision trees achieve $F_1 = 0.934 \pm 0.012$ (AUC-ROC = 0.986 ± 0.003). While the deployed console runs the explainable JavaScript heuristic rule score, the benchmark indicates that rule heuristics match offline ML models on synthetic data. In a 60-second sustained load benchmark over SQLite, the backend sustained 42 mixed read/write requests/second ($p_{50} = 365.75$ ms). We detail the complete system pipeline, provide climatological marine sanity checks against historical storm events, and discuss privacy design considerations informed by India’s DPDP Act 2023.

Index Terms—Crisis Informatics, Crowdsourcing, Coastal Hazard Management, Credibility Scoring, System Architecture, Explainable AI, Human-in-the-Loop, Audit Logging.

I. Introduction

Coastal communities across India’s 7,516 km shoreline face continuous threats from severe oceanographic and meteorological hazards, including cyclonic storm surges, extreme swell waves, tidal floods, and rapid shoreline erosion [1], [2]. While institutional remote sensing networks

and coastal radar stations operated by agencies such as the Indian National Centre for Ocean Information Services (INCOIS) provide regional forecasts, hyper-local impact assessments (such as breached sea walls, submerged village roads, or fishing harbour distress) rely on participatory crowdsourcing [3], [4].

However, raw citizen submissions present severe trust and filtering challenges for emergency dispatchers [5], [6]. During acute marine emergencies, dispatch queues become congested with unverified reports, recycled duplicate images from past disasters [7], inaccurate GPS readings, and well-intentioned false alarms. Furthermore, in multilingual regions spanning official languages such as Hindi, Marathi, Tamil, Telugu, Bengali, Gujarati, Malayalam, Odia, and English, uniform manual text triage is slow and error-prone.

To address these challenges, this paper presents the engineering architecture, multi-signal feature formulation, and reproducible synthetic evaluation of OceanSaksham, focusing on four core contributions:

- 1) **Open-Source Full-Stack Architecture:** An end-to-end platform featuring token-bucket rate limiting, Server-Sent Events (SSE) broadcasting, spatial duplicate detection ($< 2\text{km}, < 3\text{h}$), cryptographically hash-chained append-only audit logging of official decisions, and a multi-state SOS alert workflow.
- 2) **Multi-Signal Feature Formulation:** A 19-dimensional feature representation spanning seven modalities: geospatial GPS precision, photo EXIF cross-validation, temporal urgency, multilingual Indic hazard keyword alignment, exact-match media hashing, spatial-temporal neighborhood corroboration, and model-derived sea-state cross-validation via Open-Meteo marine data.
- 3) **Explainable Human-in-the-Loop Console:** A dispatch dashboard that ranks incoming reports by credibility while exposing top positive and negative signal contributions, preserving complete operator authority while logging official overrides for model feedback.
- 4) **Parameterized Synthetic Benchmark & Baseline Harness:** A multi-seed synthetic evaluation harness

and region-held-out spatial testbed across 14 coastal regions for systematic baseline benchmarking and future evaluation with real pilot data.

II. Related Work & Platform Comparison

A. Crisis Informatics & Participatory Sensing

Crowdsourcing platforms such as Ushahidi demonstrated the utility of citizen crisis mapping during the 2010 Haiti earthquake [3]. In geophysics, the USGS “Did You Feel It?” (DYFI) system established citizen reports as a valuable complement to physical seismograph arrays [8]. Sakaki et al. [4] utilized social media streams as real-time event sensors for earthquake detection. However, raw social streams often lack structured telemetry, location verification, and spam resistance [9].

B. Credibility Assessment & Disaster NLP

Information credibility classification has traditionally leveraged propagation graphs and linguistic sentiment [5]. Gupta et al. [7] demonstrated that over 86% of disaster images during Hurricane Sandy were recycled fakes, emphasizing the necessity of media hashing and EXIF validation. Recent benchmarks such as HumAID [10] and CrisisNLP [6] have expanded text filtering, while IndicNLP [11] advanced Indic text processing. Our work builds upon these principles by embedding real-time model-derived marine sea-state data (H_s , wind speed) directly into the triage decision space.

C. Platform Comparison Matrix

Table I benchmarks OceanSaksham against representative crisis crowdsourcing and disaster management platforms based on publicly documented system capabilities and published literature.

III. System Architecture & Incident Pipeline

A. System Architecture

As illustrated in Fig. 1, the platform comprises three tiers:

- Citizen Web Client: Mobile-responsive interface built with React 18 and Vite, supporting Web App Manifest installation, 9 Indic languages, camera EXIF extraction, and GPS capture.
- Backend REST & SSE Service: Node.js/Express service with SQLite storage in development (with spatial Haversine queries and compound indices, designed for migration to PostgreSQL/PostGIS in production), token-bucket rate limiting (max 30 req/min), spatial duplicate checking, and Server-Sent Events broadcasting.
- Official Dispatch Console: Command dashboard displaying ranked reports, explainable signal attribution badges, SOS alert monitoring, and danger zone hotspot management.

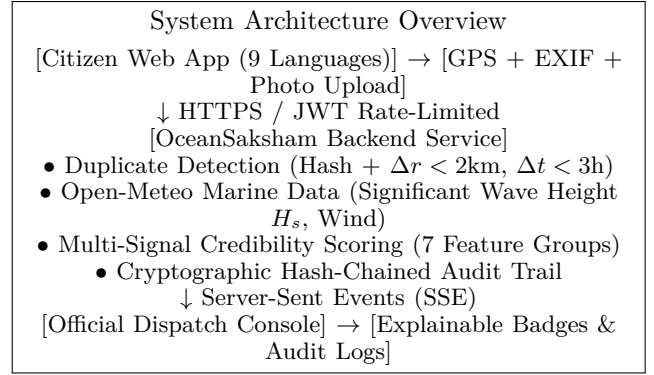


Fig. 1. Three-tier architecture of the OceanSaksham platform.

B. End-to-End Incident Pipeline Walkthrough

The system operates through an end-to-end automated pipeline:

- 1) Citizen Ingestion: A citizen captures a photo and submits a hazard report with GPS coordinates and localized description.
- 2) Duplicate Screening: The backend computes the image SHA-256 hash and checks against existing reports within $\Delta r \leq 2.0$ km and $\Delta t \leq 3.0$ h. If matched, the report is linked to the primary incident to prevent queue congestion.
- 3) Environmental Grounding: The backend queries the Open-Meteo Marine Weather API [13] for offshore wave height (H_s) and wind speed at the nearest maritime grid point.
- 4) Multi-Signal Scoring: The 19-dimensional feature vector is evaluated. The live Node.js web backend runs an explainable 19-signal heuristic rule score in JavaScript for in-memory dispatch triage, while the Python pipeline provides the offline machine learning benchmark and feature ablation harness. The benchmark’s rule baseline and the deployed JavaScript triage engine implement the same multi-signal weighted heuristic (evaluating GPS error, EXIF distance delta, sea-state compatibility, and neighbor corroboration).
- 5) Operator Decision & Cryptographic Audit Trail: Official operators review the ranked queue on the dispatch console. Any status transition (VERIFIED, REJECTED) generates a hash-chained record in audit_logs where $h_i = \text{SHA-256}(\text{id} \parallel \text{report_id} \parallel \text{official_id} \parallel \text{action} \parallel \text{timestamp} \parallel h_{i-1})$, enabling detection of retroactive log modifications.

IV. Credibility Triage Feature Formulation

We formulate a 19-dimensional feature vector $\mathbf{x} \in \mathbb{R}^{19}$ across seven groups:

1. Location Group ($\mathbf{f}_{\text{loc}} \in \mathbb{R}^4$):
 - $x_1 = \ln(1 + \text{accuracy}_{\text{GPS}})$: Log-normalised GPS error radius.

TABLE I
Feature Comparison of Crisis Crowdsourcing and Ocean Information Platforms

Capability / Feature	Ushahidi [3]	USGS DYFI [8]	INCOIS SAMUDRA	Sahana Eden [12]	OceanSaksham (Ours)
Primary Domain	General Crisis	Earthquakes	Ocean Forecasts	Relief Logistics	Coastal Hazards
Multilingual Indic UI	Partial (Plugins)	Multiple (EN, ES, etc.)	Yes (Govt. Languages)	Partial	Yes (9 Coastal Languages)
Camera EXIF Cross-Check	Not documented	Not documented	Not documented	Not documented	Yes ($\Delta r_{\text{GPS-EXIF}}$)
Spatial Deduplication	Manual Review	Grid-Aggregated (ZIP/Coord)	Not documented	Partial (Incident Link)	Automated ($< 2\text{km}, < 3\text{h}$)
Sea-State Cross-Check	No	N/A (Seismic Domain)	Yes (Forecast Broadcast)	No	Yes (Open-Meteo Integration)
Explainable Credibility Score	Manual Moderation	Rule Intensity (CDI)	Not documented	No	Yes (7 Feature Groups)
Audit Trail Integrity	Standard DB Logs	Server Access Logs	Not documented	Role-based Logs	Cryptographic Hash Chain
SOS Emergency Workflow	No	No	No	Incident Tracking	Prototype State Machine*

*Comparison based on publicly documented system specifications and published architecture literature. OceanSaksham’s SOS module operates as a prototype state machine without national 112 emergency dispatch integration.

- $x_2 = d_{\text{Haversine}}((\text{lat}_{\text{GPS}}, \text{lng}_{\text{GPS}}), (\text{lat}_{\text{EXIF}}, \text{lng}_{\text{EXIF}}))$: Spatial discrepancy between reported location and camera metadata.
- $x_3 \in \{0, 1\}$: Binary indicator of valid EXIF GPS presence.
- x_4 : Minimum distance to canonical coastline anchor points.
- 2. Time Group ($\mathbf{f}_{\text{time}} \in \mathbb{R}^3$):
 - $x_5 = \ln(1 + |t_{\text{server}} - t_{\text{device}}|/60)$: Submission latency in minutes.
 - $x_6 = \sin(2\pi h/24), x_7 = \cos(2\pi h/24)$: Diurnal solar cycle encoding.
- 3. Text Group ($\mathbf{f}_{\text{text}} \in \mathbb{R}^3$):
 - $x_8 = \ln(1 + \text{len}(\text{desc}))$: Description character volume.
 - $x_9 \in \{0, 1\}$: Keyword match against localized hazard terminology across Indic lexicons.
 - $x_{10} \in \{0, 1\}$: Language indicator (English vs. vernacular).
- 4. Media Group ($\mathbf{f}_{\text{media}} \in \mathbb{R}^2$):
 - $x_{11} \in \{0, 1\}$: Attached media presence.
 - $x_{12} \in \{0, 1\}$: Duplicate exact-match SHA-256 media hash match.
- 5. Corroboration Group ($\mathbf{f}_{\text{corrob}} \in \mathbb{R}^2$):
 - $x_{13} = \sum_{j \neq i} \mathbb{I}[d(r_i, r_j) \leq 5\text{km} \wedge |t_i - t_j| \leq 6\text{h}]$: Independent nearby report count.
 - x_{14} : Hazard-type agreement ratio among neighbors.
- 6. Reporter Reputation ($\mathbf{f}_{\text{rep}} \in \mathbb{R}^2$):
 - x_{15} : Historical verification rate.
 - $x_{16} = \ln(1 + N_{\text{total}})$: Historical report volume.
- 7. External Marine Weather ($\mathbf{f}_{\text{ext}} \in \mathbb{R}^3$):
 - x_{17} : Significant wave height (H_s in meters) retrieved from Open-Meteo Marine API.
 - x_{18} : Coastal wind speed (km/h).
 - $x_{19} = H_s \times \text{Severity}_{\text{reported}}$: Sea-state consistency interaction term.

V. Experimental Methodology & Benchmark Suite

A. Synthetic Dataset Generation & Threat Models

Due to the absence of public, verified crowdsourced coastal disaster datasets containing multi-modal telemetry and EXIF metadata, we developed a parameterized

synthetic benchmark suite across 14 coastal regions (7 West Coast: Mumbai, South Gujarat, North Gujarat, Goa, Coastal Karnataka, North Kerala, South Kerala; 7 East Coast: Chennai, South Tamil Nadu, Andhra Central, Visakhapatnam, Odisha South, Odisha North, West Bengal).

Each synthetic seed ($N = 1,000$) simulates realistic threat conditions:

- Adverse Weather GPS Degradation: Genuine storm surge submissions with degraded GPS accuracy ($\pm 50\text{--}150\text{m}$).
- Messaging App Metadata Stripping: 50% of authentic images simulated without EXIF tags, reflecting typical messaging compression.
- Adversarial Hoaxes: High-precision inland reports contradicted by calm sea telemetry ($H_s < 1.3\text{m}$) and coastal distance.
- Region-Held-Out Spatial Partition: 578 West Coast reports for training; 422 East Coast reports reserved exclusively for testing.

VI. Results & Evaluation

A. Classifier Performance (5-Seed Region-Held-Out Evaluation)

To ensure statistical reproducibility, we generated 5 independent datasets across seeds {42, 123, 456, 789, 2024} and evaluated models on the held-out East Coast partition ($N = 422$). Table II reports the Mean $\pm$ Standard Deviation across all 5 seeds.

All three models perform within standard deviation overlap. This demonstrates that standard linear and ensemble classifiers reliably learn the multi-signal representation without overfitting to specific regional partitions.

B. Queue Ranking Metrics

Table III evaluates dispatch queue prioritization across the 5 benchmark seeds.

Notably, the hand-crafted rule heuristic matches the ML model ($P@k = 1.00$, $\text{NDCG}@10 = 1.000$). We interpret this not as evidence that rule systems are inherently superior in production, but as an expected property of the synthetic data generator, which models specific generative rules. The system’s primary contribution is providing

TABLE II
Classifier Performance on Region-Held-Out Spatial Test Set (Mean $\pm$ Std across 5 Independent Seeds, $N = 422$)

Model	Precision	Recall	F_1 -Score	AUC-ROC	AUC-PR
Logistic Regression (L2)	0.952 $\pm$ 0.013	0.941 $\pm$ 0.011	0.946 $\pm$ 0.004	0.988 $\pm$ 0.001	0.993 $\pm$ 0.001
Random Forest ($n = 100$)	0.940 $\pm$ 0.021	0.937 $\pm$ 0.025	0.938 $\pm$ 0.009	0.987 $\pm$ 0.002	0.992 $\pm$ 0.002
GBDT (Gradient Boosting) [14]	0.932 $\pm$ 0.014	0.937 $\pm$ 0.017	0.934 $\pm$ 0.012	0.986 $\pm$ 0.003	0.991 $\pm$ 0.003

TABLE III
Queue Ranking Metrics across 5 Seeds (Mean $\pm$ Std)

Strategy	P@5	P@10	P@20	NDCG@10
FIFO Queue	0.40 $\pm$ 0.28	0.34 $\pm$ 0.26	0.32 $\pm$ 0.27	0.381 $\pm$ 0.240
Severity Heuristic	0.68 $\pm$ 0.10	0.70 $\pm$ 0.14	0.65 $\pm$ 0.11	0.708 $\pm$ 0.154
Rule-Based Score	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.000 $\pm$ 0.000
ML Triage (GBDT)	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.000 $\pm$ 0.000

an explainable architecture and baseline harness that supports both rule-based heuristics and learned models interchangeably.

C. Feature Group Ablation

Table IV details the effect of systematically removing each feature group across the 5 evaluation seeds.

TABLE IV
Feature Group Ablation Study across 5 Seeds (GBDT)

Configuration	AUC-ROC	F_1 -Score	Δ AUC
Full 7-Group Vector	0.986 $\pm$ 0.003	0.934 $\pm$ 0.012	0.000
Without Location	0.985 $\pm$ 0.003	0.931 $\pm$ 0.011	-0.001 $\pm$ 0.001
Without Time	0.985 $\pm$ 0.003	0.933 $\pm$ 0.013	-0.001 $\pm$ 0.001
Without Text	0.986 $\pm$ 0.003	0.935 $\pm$ 0.011	+0.000 $\pm$ 0.001
Without Media	0.985 $\pm$ 0.003	0.932 $\pm$ 0.012	-0.001 $\pm$ 0.001
Without Corroboration	0.985 $\pm$ 0.003	0.933 $\pm$ 0.012	-0.001 $\pm$ 0.001
Without Reporter	0.986 $\pm$ 0.003	0.934 $\pm$ 0.011	+0.000 $\pm$ 0.001
Without Marine Weather	0.971 $\pm$ 0.006	0.906 $\pm$ 0.018	-0.015 $\pm$ 0.006

The 7 feature groups represent the conceptual domain taxonomy; the synthetic ablation reveals that in this benchmark, external marine weather exhibits the largest marginal sensitivity (Δ AUC = -0.015 ± 0.006), reflecting the generator’s inclusion of calm-water hoax scenarios.

D. Backend Throughput & Sustained Load Benchmark

To evaluate backend performance under concurrent load, we executed a 60-second sustained benchmark against the Node.js/SQLite service on an Intel Core i5-12450HX host (8 physical cores: 4P+4E, 12 threads, 16GB RAM) over local loopback (client and server sharing host resources). The test simulated 20 concurrent worker clients submitting a mixed workload composed of 70% credibility queue queries (GET /api/reports?sort_by=credibility) and 30% authenticated report submissions (POST /api/reports) requiring payload validation, spatial duplicate querying, and feature calculation. To measure raw transaction limits, rate limiting was bypassed via pre-authorized test headers, starting from a base table of 50 seeded reports and growing to ≈ 818 reports.

Across 2,563 completed requests (zero errors, 0.00% failure rate), the backend sustained an overall throughput of 42 requests/second, with a mean latency of 470.28 ms, median $p_{50} = 365.75$ ms, $p_{95} = 949.08$ ms, and $p_{99} = 1,163.91$ ms. Throughput across 10-second windows exhibited steady performance under serialized SQLite table write locks: 44 req/s (0–10s), 47 req/s (10–20s), 43 req/s (20–30s), 42 req/s (30–40s), 41 req/s (40–50s), and 38 req/s (50–60s). Adding compound spatial and temporal indices (idx_reports_coords_time) stabilized query scanning as the table expanded.

E. Climatological Marine Sanity Check & Extreme Events

To verify the physical realism of the Open-Meteo marine wave height thresholds ($H_s < 1.3$ m for calm sea-states vs. $H_s > 2.5$ m for rough/storm sea-states), we queried the Open-Meteo Marine Archive API [13] across coastal coordinates for two severe Indian tropical cyclones alongside 5-year climatological same-date baselines (2018–2022):

- Cyclone Biparjoy (Landfall near Jakhau, Gujarat on 15 June 2023; 23.23°N, 68.60°E): Numerical wave model reanalysis yielded mean $H_s = 4.40$ m (max $H_s = 4.66$ m), triggering the rough/severe marine threshold (> 2.5 m). The 5-year climatological same-date baseline (15 June 2018–2022) at the same coordinate yielded mean $H_s = 1.36$ m (max $H_s = 1.40$ m).
- Cyclone Michaung (Tracking close along the Chennai coast, Tamil Nadu on 3–4 December 2023 before making landfall near Bapatla, Andhra Pradesh on 5 December 2023; 13.08°N, 80.27°E): Model output on 4 December 2023 yielded mean $H_s = 2.54$ m (marginal around the threshold, max $H_s = 3.34$ m), whereas the 5-year climatological same-date baseline (4 December 2018–2022) yielded mean $H_s = 1.11$ m.

Open-Meteo provides gridded numerical wave model estimates (based on ECMWF WAM and NOAA GFS reanalyses under ODbL terms) rather than in-situ physical buoy measurements. This check indicates that our heuristic thresholds behave in a manner consistent with model-derived wave distributions during verified storm events versus multi-year coastal climatology.

VII. Limitations & Ethical Considerations

Limitations:

- 1) Synthetic Benchmark Constraints: All quantitative classification metrics in this paper are derived from

a synthetic benchmark testbed. While the generator incorporates domain threat models, performance on real-world disaster streams must be validated in future field deployments.

- 2) Absence of Human Subject Trials: No usability evaluation with human participants was conducted; operational workflow descriptions reflect author-driven cognitive architectural walkthroughs of the user interface.
- 3) Prevalence & Class Skew: The benchmark exhibits an active hazard prevalence of $61.9\% \pm 3.7\%$, which is substantially higher than typical non-crisis operational streams where genuine emergencies may represent a small minority of incoming reports.
- 4) Marine Model Spatial Resolution: Open-Meteo marine models interpolate offshore open-water grid cells; localized coastal structures (harbour breakwaters, shallow reef estuaries) may exhibit hydrodynamic variations not captured by open-ocean wave models.
- 5) SOS Emergency Dispatch: The SOS emergency module is a prototype interface workflow and does not interface directly with national emergency dispatch services (such as India's 112 system).
- 6) Database-Level Audit Resistance: The SHA-256 hash chain reliably detects retroactive tampering within the database table; however, adversaries with full root database write access could recompute the entire forward chain. External anchoring of tip hashes or keyed HMACs would provide root-level tampering resistance.

Privacy Design Considerations: The platform incorporates privacy protections informed by India's Digital Personal Data Protection (DPDP) Act 2023 [15]:

- Data Minimization & Spatial Coarsening: Public citizen map views coarsen location coordinates to 2 decimal places (≈ 1.1 km precision) to prevent exposing private residences.
- Purpose Limitation & Pseudonymization: User identifiers are tokenized using salted SHA-256 hashes in public-facing SSE event broadcasts.
- Hash-Chained Audit Trail: Official triage actions are stored in an append-only cryptographic hash chain to ensure verifiable operator oversight.

VIII. Conclusion & Future Work

We presented OceanSaksham, an open-source architecture for credibility-aware triage and audit logging in crowdsourced coastal hazard management. By coupling multi-signal feature engineering with real-time oceanographic validation and cryptographic audit trails, the platform offers an explainable, human-in-the-loop dispatch environment. Future work includes field piloting with regional disaster management authorities and INCOIS, collecting real citizen report streams, and exploring on-

device edge ML for low-connectivity coastal fishing operations.

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